Lecture 6: Generalized Functions and Fourier Transforms

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Dirac delta function
- Step function (Heaviside step function)
- Rect function
- Fourier transforms

Response to Objects

- In optical imaging systems, when a small bright light source is much smaller than the spatial resolution capability of the system, the object is essentially a **mathematical point**, referred to as a **point source**. The resulting image or measured data represents the system's **point spread function (PSF)**.
- A more general light sources can be considered as a **collection of points**.
- If the system is **linear**, the response to the general object can be obtained by adding up the responses to the individual points.
- The mathematical construct that corresponds to a **physical point source** is the **Dirac delta function**.

Well-behaved Functions

A function is considered **well-behaved** if it is:

- Single-valued,
- Continuous, and
- Differentiable.

The Dirac delta function **does not** satisfy these conditions. It is *neither continuous nor differentiable* in the classical sense. This motivates the need for a *wider class of functionals*: **distributions**.

Distributions

A **distribution** is a **linear**, **continuous *functional*** that maps a function to a real or complex number. Distributions do not require the kernel to belong to Hilbert space.

Function, Functional, and Integral Transform

- **Function**: A mapping from a *scalar* to a *scalar*
  - Example: If x is a real value, a linear function $y = f(x)$ maps $\mathbb{R}^1$ to $\mathbb{R}^1$.

- **Functional**: A mapping from a *function* to a *scalar*
  - Example: If $f(x)$ is a square-integrable function, a functional $\Phi[f(x)]$ maps $\mathbb{L}_2$ to $\mathbb{C}^1$.
  - General form of a linear functional on a Hilbert space: $\Phi[f(x)] = \langle \mathbf{h}, \mathbf{f} \rangle = \int_{-\infty}^{\infty} dx\, h^*(x)\, f(x)$
- **Integral transform (continuous-to-continous mapping)**: A mapping from a *function* to a *function*
  - General form: $g(x') = \int_{\alpha}^{\beta} dx\, h(x', x)\, f(x)$ maps $\mathbb{L}_2$ to $\mathbb{L}_2$.

We use a wider class of functionals, i.e., **distributions**, relaxing the requirement that the kernel lie in a Hilbert space.

In functionals:

$$\Phi[f(x)] = \int_{-\infty}^{\infty} dx\, h(x)\, f(x),$$

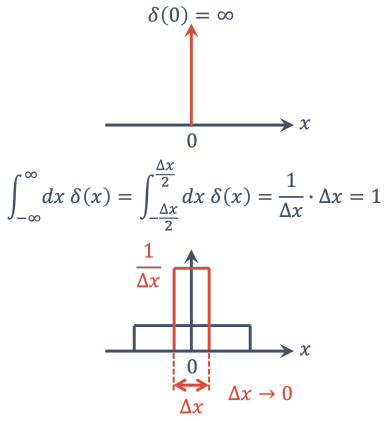
the kernel $h(x)$ does *not* need to be *square-integrable*. Instead, we *specify the mapping rule* of the functional, treating the kernel function as a **generalized function**.

Dirac Delta Function

Mathmetically, the **Dirac delta function** is a **distribution**, not a conventional function. By *specifying the mapping rule* of the functional, the kernel function becomes a **generalized function**:

$$\delta(x) = \begin{cases} \infty & \text{if } x = 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$\int_{-\infty}^{\infty} dx\, \delta(x) = 1.$$



$$\delta(x - x_0) = \begin{cases} \infty & \text{if } x = x_0, \\ 0 & \text{otherwise,} \end{cases}$$

$$\int_{-\infty}^{\infty} dx\, \delta(x - x_0) = 1.$$

Example Uses

$$\int_{\alpha}^{\beta} dx\, \delta(x - x_0)\, f(x)$$

$$\int_{-\infty}^{\infty} dx\, \delta\left(x - \frac{\pi}{2}\right) \sin(x^2) = \sin\left(\frac{\pi^2}{4}\right)$$

If the kernel of $\mathcal{H}$ is $h(x) = \delta\left(x - \frac{\pi}{2}\right)$ and $f(x) = \sin(x^2)$, the operator $\mathcal{H}$ **samples** a value from $f(x)$.

📝 Discrete Implementation of the Example

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt

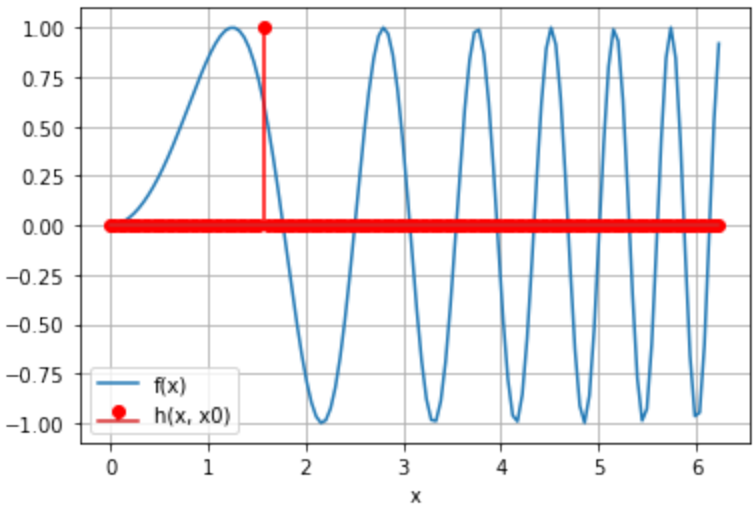
# Define object f(x)
def f(x):
  return np.sin(x**2)

# Define kernel function, discrete Dirac delta function
def h(x, x0):
  return np.where(x==x0, 1, 0)

# Define the sample points
xs = np.arange(0, 2*np.pi, np.pi/64)
x0 = np.pi/2.
```

In [2]:
```
# Plot f(x) and h(x, x0)
plt.plot(xs, f(xs), label="f(x)")
plt.stem(xs, h(xs, x0), linefmt="r", label="h(x, x0)")
plt.xlabel("x")
```

```
plt.legend()
plt.grid()
plt.show()
```



In [3]:
```
# Calculate expected result
g_exp = np.sin(np.pi**2/4)

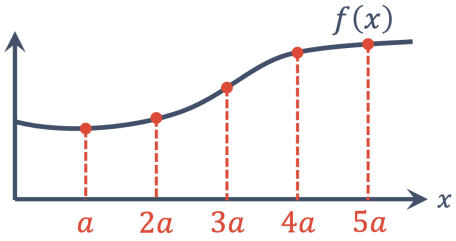
# Calculate result
g = np.sum(h(xs, x0)*f(xs))

# Print results
print(f"Expected result: {g_exp}")
print(f"Result:          {g}")
```

```
Expected result: 0.6242659526396992
Result:          0.6242659526396992
```

Sampling with an interval a:

$$\int_{-\infty}^{\infty} dx\, \delta(x - n \cdot a)\, f(x), \quad n \in \mathbb{Z}.$$



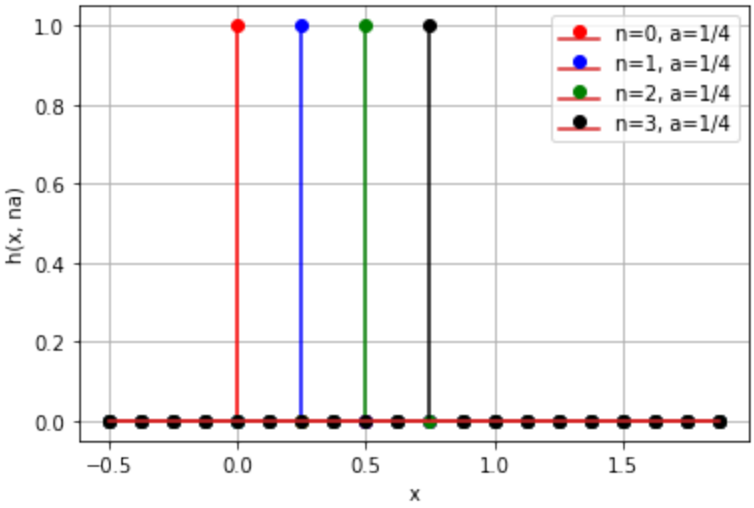
📝 Plotting Discrete Implementation of the Example

In [4]:
```
# Define kernel function, discrete Dirac delta function
def h(x, na):
  return np.where(x==na, 1, 0)

# Define the sample points
xs = np.arange(-0.5, 2., 1/8)

# Define the kernel parameters
ns = np.arange(0, 5) # Integer index
a = 1/4. # Interval
nas = ns*a
```

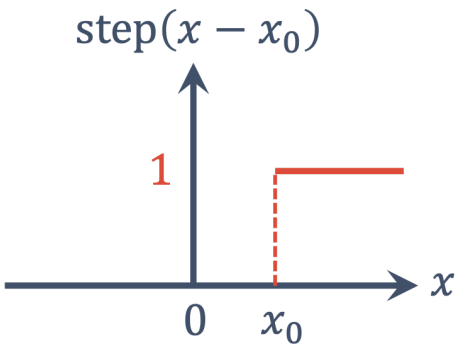
In [5]:
```
# Plot h(x, na)
plt.stem(xs, h(xs, nas[0]), linefmt="r", label="n=0, a=1/4")
plt.stem(xs, h(xs, nas[1]), linefmt="b", label="n=1, a=1/4")
plt.stem(xs, h(xs, nas[2]), linefmt="g", label="n=2, a=1/4")
plt.stem(xs, h(xs, nas[3]), linefmt="k", label="n=3, a=1/4")
plt.xlabel("x")
plt.ylabel("h(x, na)")
plt.legend()
plt.grid()
plt.show()
```



➡️ The Dirac delta function is useful in **continuous-to-discrete (C-D) mappings**.

Step Function (Heaviside Step Function)

$$\text{step}(x - x_0) = \begin{cases} 1 & \text{if } x > x_0, \\ 0 & \text{if } x < x_0. \end{cases}$$



Example Use

$$\int_{\alpha}^{\beta} dx\, \text{step}(x - x_0)\, f(x)$$

$$\int_{-\infty}^{\infty} dx\, \text{step}\left(x - \frac{1}{2}\right) x^2 e^{-3x} = \int_{\frac{1}{2}}^{\infty} dx\, x^2 e^{-3x}$$

If the kernel of $\mathcal{H}$ is $h(x) = \text{step}(x - x_0)$, the $\mathcal{H}$ acts as a **switch**.

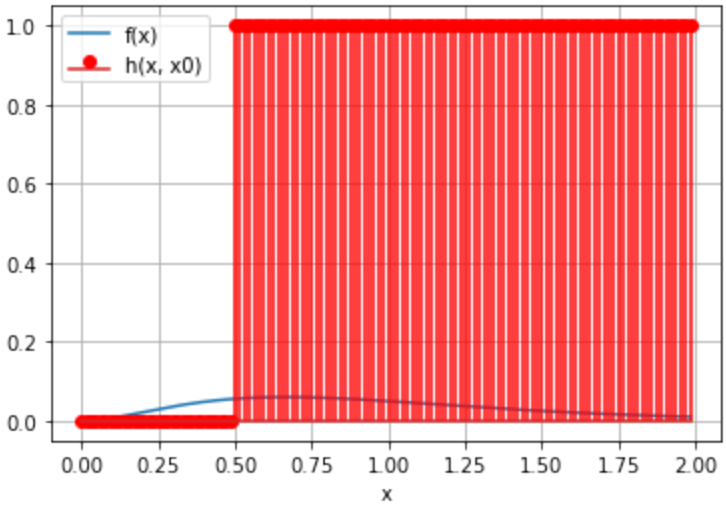
📝 Plotting Discrete Implementation of the Example

In [6]:
```
# Define object f(x)
def f(x):
  return x**2*np.exp(-3*x)

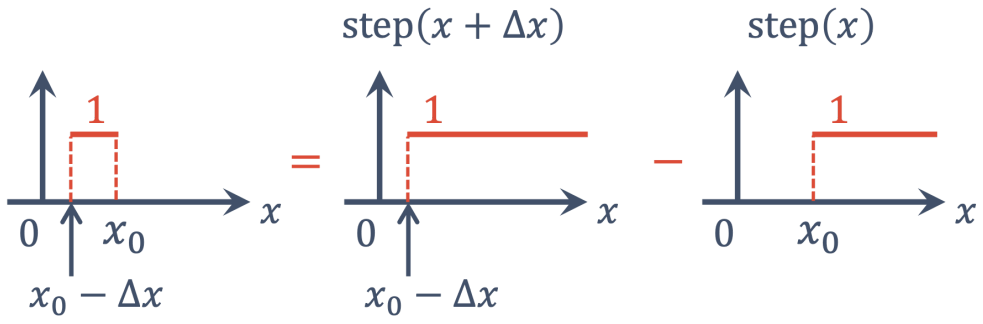
# Define kernel function, discrete step function
def h(x, x0):
  return np.where(x>=x0, 1, 0)

# Define the sample points
xs = np.arange(0, 2., 1/64)
x0 = 1/2.
```

In [7]:
```
# Plot f(x) and h(x, x0)
plt.plot(xs, f(xs), label="f(x)")
plt.stem(xs, h(xs, x0), linefmt="r", label="h(x, x0)")
plt.xlabel("x")
plt.legend()
plt.grid()
plt.show()
```



Derivative of the Step Function and its Connection to the Delta Function



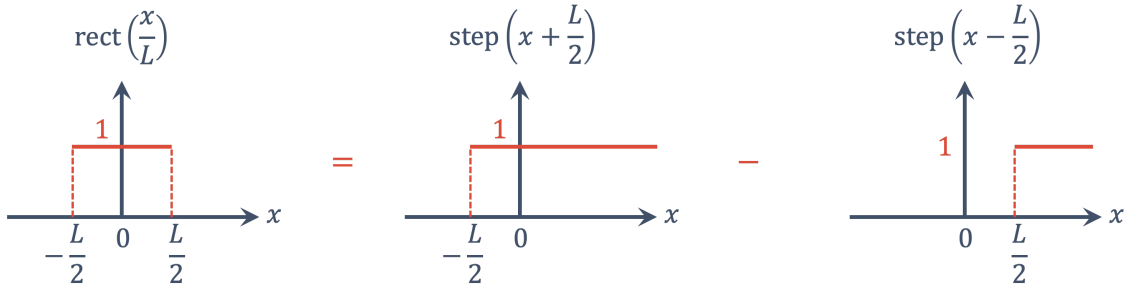
$$\frac{d}{dx}\text{step}(x - x_0) \quad \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad \text{Derivative of } f(x)$$

$$= \begin{cases} \lim\limits_{\Delta x \to 0} \dfrac{1}{\Delta x} & \text{if } x_0 - \Delta x < x < x_0, \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \infty & \text{if } x = x_0, \\ 0 & \text{otherwise} \end{cases} \quad = \delta(x - x_0)$$

Rect Function

$$\text{rect}\left(\frac{x}{L}\right) = \begin{cases} 1 & \text{if } |x| < \dfrac{L}{2}, \\ 0 & \text{if } |x| > \dfrac{L}{2}, \end{cases} \quad \text{rect}\left(\frac{x}{L}\right) = \text{step}\left(x + \frac{L}{2}\right) - \text{step}\left(x - \frac{L}{2}\right)$$

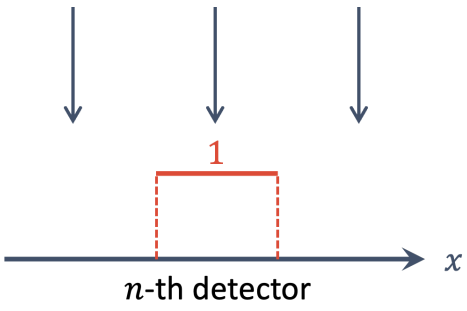


The rect function can be seen as a **window** function that selects a specific inverval of a function.

Example Use

Sampling with an interval a and **a width of** L:

$$\int_{-\infty}^{\infty} dx \, \text{rect}\left(\frac{x - n \cdot a}{L}\right) f(x), \quad n \in \mathbb{Z}.$$



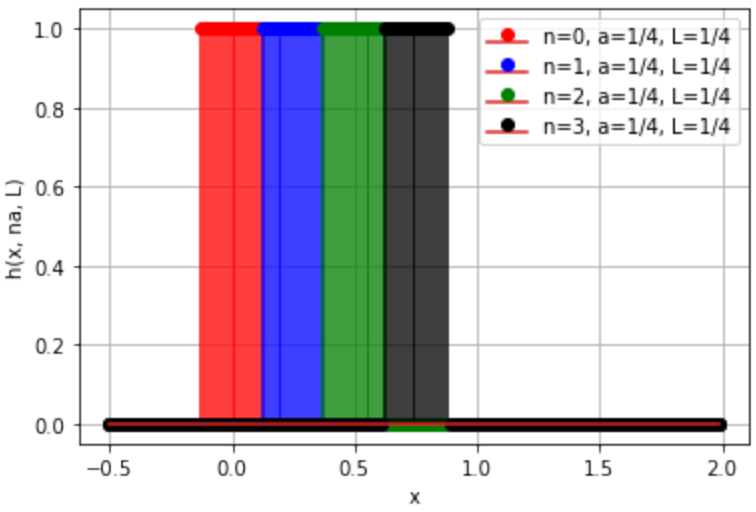
📝 Plotting Discrete Implementation of the Example

In [8]:
```python
# Define kernel function, discrete rect function
def h(x, na, L):
  return np.where((x>=na-L/2) & (x<=na+L/2), 1, 0)

# Define the sample points
xs = np.arange(-0.5, 2., 1/64)

# Define the kernel parameters
ns = np.arange(0, 4) # Integer index
a = 1/4. # Interval
nas = ns*a
L = 1/4. # Width
```

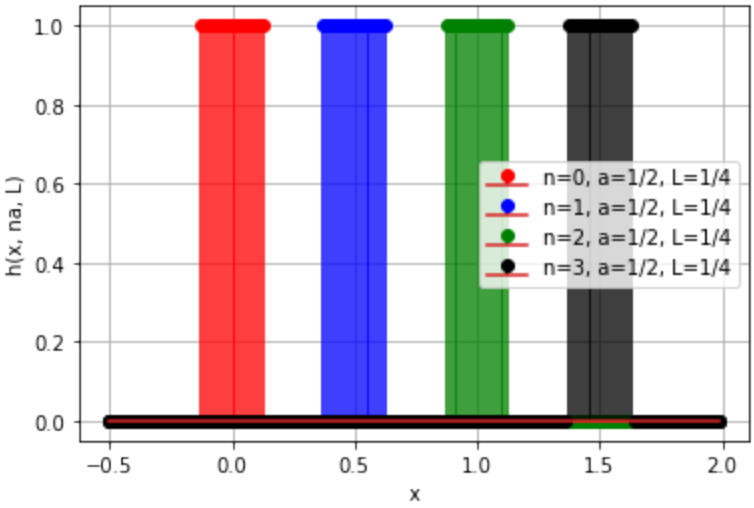
In [9]:
```python
# Plot h(x, na, L)
plt.stem(xs, h(xs, nas[0], L), linefmt="r", label="n=0, a=1/4, L=1/4")
plt.stem(xs, h(xs, nas[1], L), linefmt="b", label="n=1, a=1/4, L=1/4")
plt.stem(xs, h(xs, nas[2], L), linefmt="g", label="n=2, a=1/4, L=1/4")
plt.stem(xs, h(xs, nas[3], L), linefmt="k", label="n=3, a=1/4, L=1/4")
plt.xlabel("x")
plt.ylabel("h(x, na, L)")
plt.legend()
plt.grid()
plt.show()
```



In [10]:
```python
# Define the kernel parameters
ns = np.arange(0, 4) # Integer index
a = 1/2. # Interval
nas = ns*a
L = 1/4. # Width

# Plot h(x, na, L)
plt.stem(xs, h(xs, nas[0], L), linefmt="r", label="n=0, a=1/2, L=1/4")
plt.stem(xs, h(xs, nas[1], L), linefmt="b", label="n=1, a=1/2, L=1/4")
plt.stem(xs, h(xs, nas[2], L), linefmt="g", label="n=2, a=1/2, L=1/4")
plt.stem(xs, h(xs, nas[3], L), linefmt="k", label="n=3, a=1/2, L=1/4")
plt.xlabel("x")
```

```
plt.ylabel("h(x, na, L)")
plt.legend()
plt.grid()
plt.show()
```



Fourier Transforms

Fourier transform:

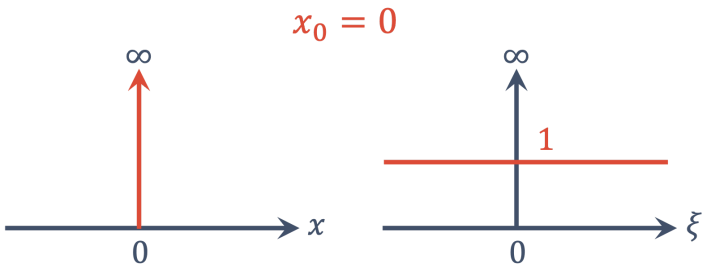
$$\mathcal{F}[f(x)] = \int_{-\infty}^{\infty} dx\, f(x)\, e^{-2\pi i \xi x}$$

Inverse Fourier transform:

$$\mathcal{F}^{-1}[F(\xi)] = \int_{-\infty}^{\infty} d\xi\, F(\xi)\, e^{2\pi i \xi x}$$

Fourier Transform of the Dirac Delta Function

$$\mathcal{F}[\delta(x - x_0)] = \int_{-\infty}^{\infty} dx\, \delta(x - x_0)\, e^{-2\pi i \xi x} = e^{-2\pi i \xi x_0}$$



Fourier Transform of the Rect Function

$$\mathcal{F}\left[\text{rect}\left(\frac{x}{L}\right)\right] = \int_{-\infty}^{\infty} dx\, \text{rect}\left(\frac{x}{L}\right) e^{-2\pi i \xi x}$$

$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} dx\, e^{-2\pi i \xi x} \quad e^{-i\theta} = \cos\theta - i\sin\theta$$

$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} dx\, [\cos(2\pi\xi x) - i\sin(2\pi\xi x)]$$

$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \cos(2\pi\xi x)$$

$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \cos(2\pi\xi x)$$

$$u = 2\pi\xi x$$

$$du = 2\pi\xi\, dx \quad dx = \frac{1}{2\pi\xi}\, du$$

$$x \to \frac{L}{2} \quad u \to 2\pi\xi\frac{L}{2} = \pi\xi L$$

$$= \int_{-\pi\xi L}^{\pi\xi L} \frac{1}{2\pi\xi}\, du \cos(u)$$

$$= \frac{1}{2\pi\xi} \int_{-\pi\xi L}^{\pi\xi L} du \cos(u)$$

$$= \frac{2}{2\pi\xi} \int_{0}^{\pi\xi L} du \cos(u)$$

$$= \frac{1}{\pi\xi}\sin(\pi\xi L) \quad \text{sinc}(x) = \frac{\sin x}{x}$$

$$= \frac{L}{\pi\xi L}\sin(\pi\xi L) = L\,\text{sinc}(\pi\xi L)$$

Similarity Theorem

$$\mathcal{F}[f(\alpha x)] = \frac{1}{|\alpha|} F\left(\frac{\xi}{\alpha}\right)$$

Rayleigh's Theorem (Energy Conservation)

$$\int_{-\infty}^{\infty} dx\, |f(x)|^2 = \int_{-\infty}^{\infty} d\xi\, |F(\xi)|^2$$

Shift Theorem

$$\mathcal{F}^{-1}[F(\xi - \xi_0)] = \int_{-\infty}^{\infty} d\xi\, F(\xi - \xi_0)\, e^{2\pi i \xi x}$$

$$= \int_{-\infty}^{\infty} d\xi\, F(\xi - \xi_0)\, e^{2\pi i \xi x}\, e^{-2\pi i \xi_0 x}\, e^{2\pi i \xi_0 x}$$

$$= e^{2\pi i \xi_0 x} \int_{-\infty}^{\infty} d\xi\, F(\xi - \xi_0)\, e^{2\pi i (\xi - \xi_0) x}$$

$$u = (\xi - \xi_0) \quad du = d\xi$$

$$= e^{2\pi i \xi_0 x} \int_{-\infty}^{\infty} du\, F(u)\, e^{2\pi i u x}$$

$$= e^{2\pi i \xi_0 x}\, f(x)$$

Convolution Theorem

$$\mathcal{F}^{-1}[F(\xi) * H(\xi)] = \int_{-\infty}^{\infty} d\xi\, F(\xi) * H(\xi)\, e^{2\pi \xi x}$$

$$F(t) * H(t) = \int_{-\infty}^{\infty} d\tau\, F(\tau) H(t - \tau)$$

$$= \int_{-\infty}^{\infty} d\xi \int_{-\infty}^{\infty} d\tau\, F(\tau) H(\xi - \tau)\, e^{2\pi \xi x}$$

$$= \int_{-\infty}^{\infty} d\tau\, F(\tau) \int_{-\infty}^{\infty} d\xi\, H(\xi - \tau)\, e^{2\pi \xi x}$$

$$= \int_{-\infty}^{\infty} d\tau\, F(\tau) \int_{-\infty}^{\infty} d\xi\, H(\xi - \tau)\, e^{2\pi \xi x}\, e^{-2\pi \tau x}\, e^{2\pi \tau x}$$

$$= \int_{-\infty}^{\infty} d\tau\, F(\tau)\, e^{2\pi \tau x} \int_{-\infty}^{\infty} d\xi\, H(\xi - \tau)\, e^{2\pi (\xi - \tau) x}$$

$$u = (\xi - \tau) \quad du = d\xi$$

$$= \int_{-\infty}^{\infty} d\tau\, F(\tau)\, e^{2\pi \tau x} \int_{-\infty}^{\infty} du\, H(u)\, e^{2\pi u x}$$

$$= \int_{-\infty}^{\infty} d\tau\, F(\tau)\, e^{2\pi \tau x} h(x) = f(x) h(x)$$